VWCE.DE Investment Forecast

Econometric Analysis of a Monthly €1,000 DCA Programme

Interactive Brokers · €3 fee/trade · VWCE TER 0.22% p.a. · Start: September 2027

Results Summary (10,000 Monte Carlo paths)

Horizon	Capital	Median Portfolio	P(loss)	XIRR	Multiple
1 year	€12,000	€12,699	27.7%	12.32%	1.06×
3 years	€36,000	€42,104	18.7%	11.28%	1.17×
5 years	€60,000	€77,663	14.0%	10.95%	1.29×
10 years	€120,000	€200,133	7.9%	10.30%	1.67×
20 years	€240,000	€696,587	3.5%	9.76%	2.90×
30 years	€360,000	€1,913,813	2.0%	9.51%	5.32×

Prepared: 18 June 2026 | Data: VWCE.DE + VT (Jun 2008–Jun 2026, 4,540 obs) |
GJR-GARCH(1,2)-t + Merton jump-diffusion + Bayesian drift

Contents

1. Asset and Data

1.1. VWCE.DE

Vanguard FTSE All-World UCITS ETF (VWCE.DE, Xetra) is an EUR-denominated accumulating ETF tracking the FTSE All-World Index (≈3,700 equities across 49 markets). AUM: ≈€66 billion. TER: 0.22% p.a., continuously deducted from NAV and modelled as a daily drag of 0.22%/252 in all simulations. Latest price: €165.46 (18 June 2026), +202.7% since inception at €54.66 in July 2019.

1.2. Extended Calibration Sample (2008–2026)

VWCE has traded since July 2019. Seven years of data excludes the GFC and several other major corrections that are essential for calibrating long-run volatility. We splice in Vanguard VT (ARCA, USD), which tracks the same FTSE All-World Index since June 2008, converting to EUR at daily EUR/USD spot rates and rescaling by 1.203 at the splice point. The resulting sample spans June 2008 to June 2026 (4,540 observations).

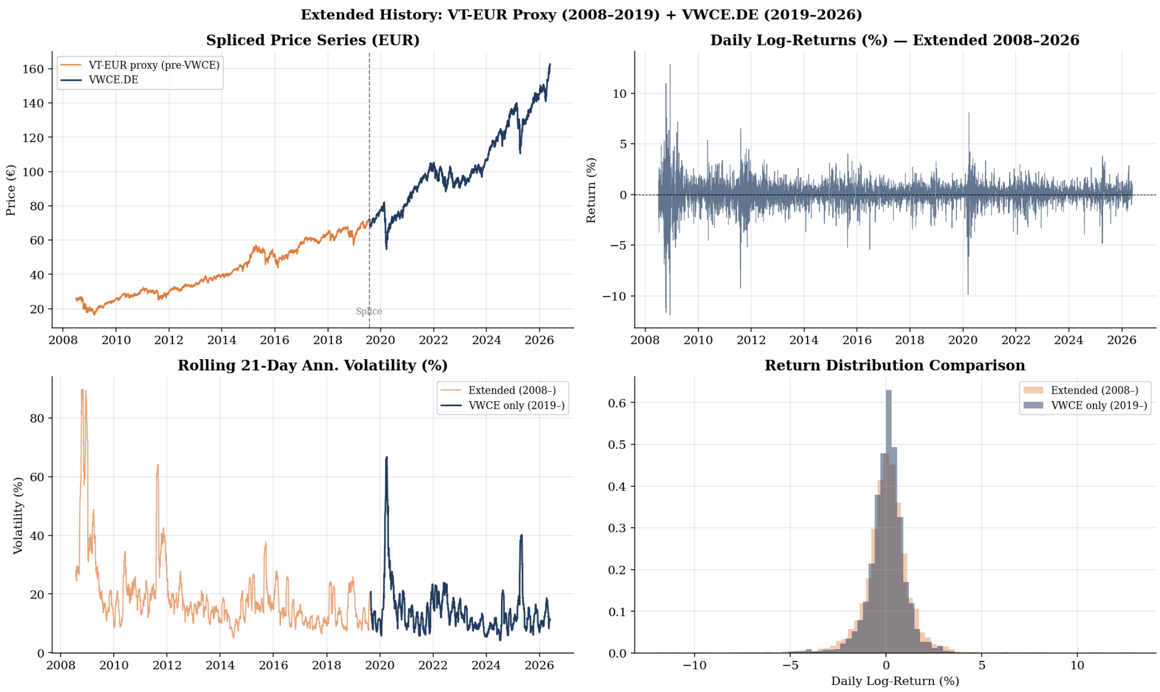

Figure 1. VT-EUR proxy (2008–2019, orange) spliced with VWCE.DE (2019–2026, blue). The GFC drives rolling volatility above 80% annualised, far beyond anything in the 7-year VWCE window. The extended return distribution has heavier tails and a wider body.

Table 1. Descriptive Statistics: Extended vs VWCE-only Sample

Statistic	Extended (2008–2026)	VWCE only (2019–2026)
Observations	4,539	1,750
Annualised return	10.28%	11.84%
Annualised volatility	20.35%	16.29%
Sharpe ratio	0.505	0.727
Skewness	−0.580	−1.062
Excess kurtosis	12.955	11.525

1.3. Exploratory Analysis

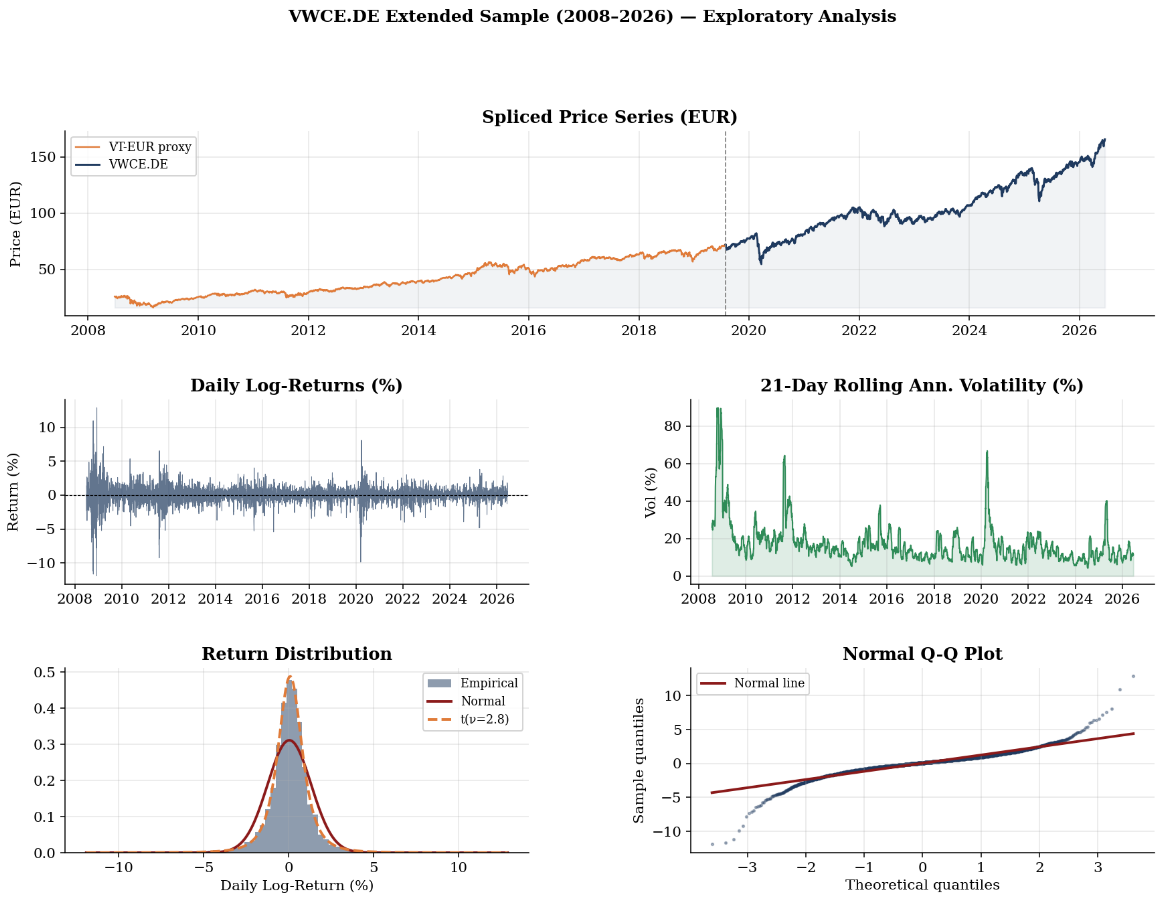

Figure 2. Exploratory analysis of the extended sample. *Top*: spliced price series (orange = VT-EUR; blue = VWCE.DE). *Middle*: daily log-returns and rolling 21-day annualised volatility. *Bottom left*: empirical distribution vs Normal and Student-t fits — fat tails are clearly visible. *Bottom right*: Normal Q-Q plot confirms extreme leptokurtosis.

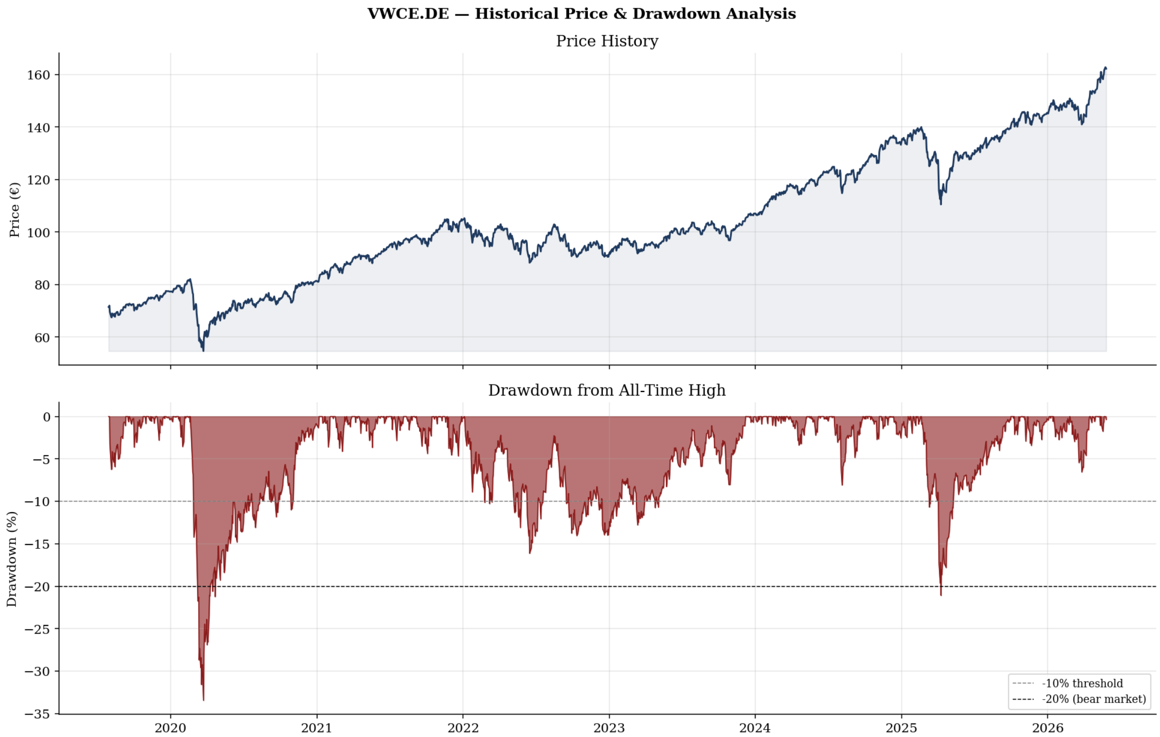

Figure 3. VWCE.DE price history and drawdown from all-time high. Three drawdown episodes: COVID (−33.4%), 2022 rate-shock (≈ −20%), 2025 turbulence (≈ −15%). All recovered. Current drawdown: −0.4%.

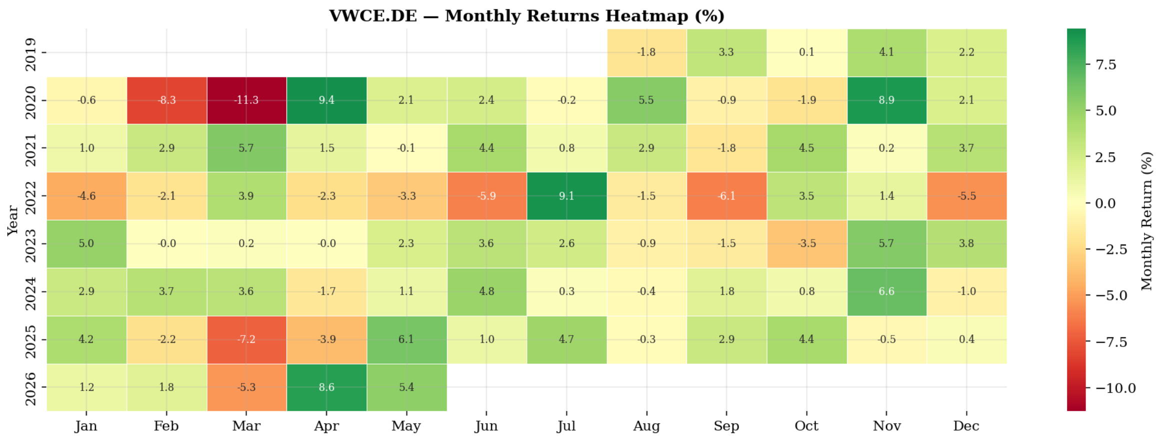

Figure 4. VWCE.DE monthly returns heatmap (2019–2026). COVID (Feb/Mar 2020: −8.3%/−11.3%), 2022 bear market (persistent red), 2026 YTD recovery (+8.6%/+5.4%). No persistent monthly seasonality.

2. Econometric Model

2.1. Pre-Testing

Table 2. Unit Root Tests and ARCH Diagnostics

Series	ADF stat	p-value	Conclusion
Log-price	+0.450	0.983	Non-stationary
Log-return	−9.823	<0.001	**Stationary**
Squared log-return	−5.374	<0.001	**Stationary**
Ljung-Box on squared returns: LB(5) = 507, LB(10) = 1,003, LB(20) = 1,278 ($p < 0.001$)			

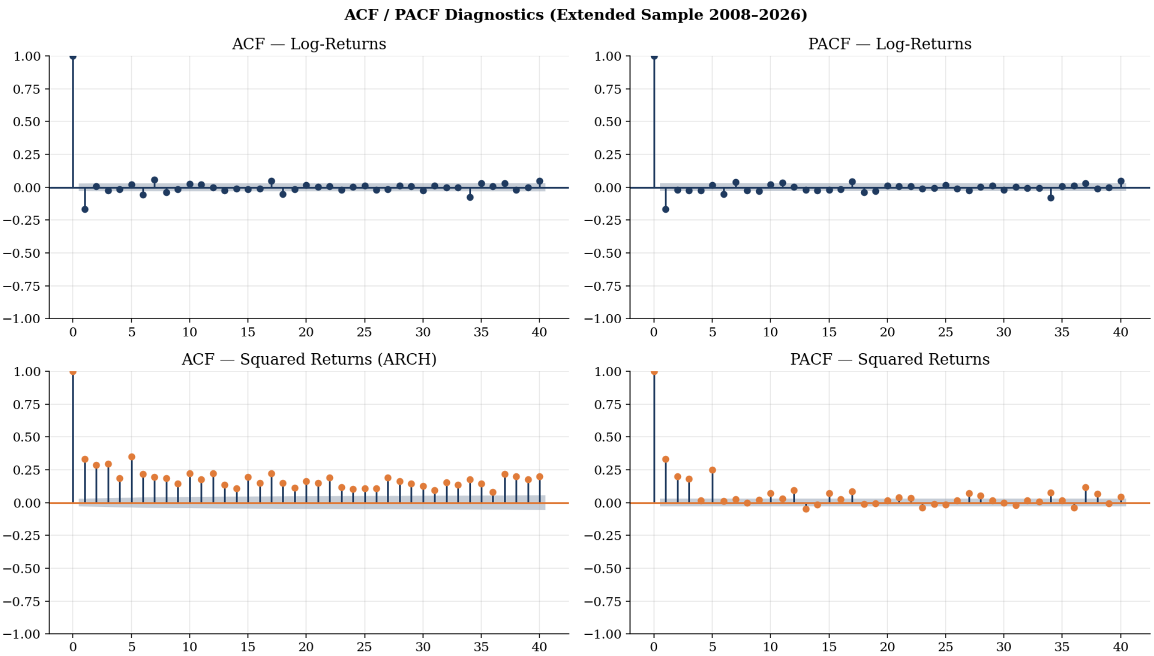

Figure 5. ACF and PACF of log-returns (top) and squared log-returns (bottom) on the extended sample. Returns show negligible autocorrelation. Squared returns exhibit large, slowly decaying autocorrelation beyond 40 lags. The PACF of squared returns cuts off after 1–3 lags, motivating GARCH(1,1) or (1,2).

2.2. Model Selection

We fit five competing GARCH specifications by MLE on the extended sample and select by AIC and BIC.

Table 3. GARCH Model Selection (extended sample, $n = 4{,}538$)

Model	AIC	BIC	Log-lik
GARCH(1,1)-Normal	12,945.4	12,977.5	−6,467.7
GARCH(1,1)-t	12,695.3	12,733.8	−6,341.6
GJR-GARCH(1,1)-t	12,612.0	12,656.9	−6,299.0
EGARCH(1,1)-t	12,710.8	12,749.3	−6,349.4
GJR-GARCH(1,2)-t	**12,608.9**	**12,660.3**	**−6,296.4**

GJR-GARCH(1,2)-t is selected by both AIC (ΔAIC = 3.1 over GJR-GARCH(1,1)-t) and

BIC. The second GARCH lag β_2 is statistically significant ($t = 2.97$, $p = 0.003$), capturing a slower secondary component of volatility mean-reversion alongside the primary β_1.

2.3. Model Specification

We combine **GJR-GARCH(1,2)** [Glosten et al., 1993] with **Student-t innovations** and a **Merton jump-diffusion** [Merton, 1976]:

Mean.

$$r_t = \mu + \phi\, r_{t-1} + \varepsilon_t + J_t, \qquad \varepsilon_t = \sigma_t\, z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} t_\nu(0,1) \tag{1}$$

Variance (GJR-GARCH(1,2)).

$$\sigma_t^2 = \omega + \left(\alpha + \gamma\, \mathbf{1}[\varepsilon_{t-1} < 0]\right) \varepsilon_{t-1}^2 + \beta_1\, \sigma_{t-1}^2 + \beta_2\, \sigma_{t-2}^2 \tag{2}$$

The leverage term γ captures asymmetric response: negative shocks raise future variance by $\gamma\varepsilon_{t-1}^2$ more than equal positive shocks. Covariance stationarity requires $\alpha + \frac{1}{2}\gamma + \beta_1 + \beta_2 < 1$.

Jump component.

$$J_t = \sum_{k=1}^{N_t} Y_k, \qquad N_t \sim \text{Poisson}(\lambda), \qquad Y_k \sim \mathcal{N}(\mu_J, \sigma_J^2) \tag{3}$$

Jumps are filtered empirically from standardised residuals with $|z| > 3.5\sigma$.

2.4. Parameter Estimates and Bayesian Drift

Table 4. GJR-GARCH(1,2)-t + Merton Jump + Bayesian Drift (extended sample, $n = 4{,}538$)

Parameter	**Est.**	**SE**	t	p	**Role**
Mean equation					
μ	0.0768	0.0123	6.24	<0.001	Daily drift (%, before shrinkage)
ϕ	−0.111	0.016	−6.87	<0.001	AR(1) mean-reversion
Variance equation (GJR-GARCH(1,2))					
ω	0.03441	0.00825	4.17	<0.001	Variance floor
α	0.0235	0.0113	2.08	0.038	Symmetric ARCH
$\boldsymbol{\gamma}$	**0.2142**	0.0345	**6.22**	**<0.001**	**Leverage (asymmetry)**
$\boldsymbol{\beta_1}$	**0.5681**	0.0934	**6.08**	**<0.001**	**Primary GARCH lag**
$\boldsymbol{\beta_2}$	**0.2712**	0.0912	**2.97**	**0.003**	**Secondary GARCH lag**
$\boldsymbol{\nu}$	**6.31**	0.55	**11.4**	**<0.001**	**Student-t d.f.**
Persistence $\alpha + \frac{1}{2}\gamma + \beta_1 + \beta_2$	0.9699				Half-life ≈23 days
Uncond. vol	16.98% p.a.				
Jump component (filtered at $\|z\| > 3.5\sigma$; 22 events in 4,539 obs)					
λ	1.22/yr				Poisson arrival rate
μ_J	−3.21%				Mean jump size (crashes)
σ_J	1.83%				Jump size std dev
Bayesian drift (prior: $\mu_0 = 7\%$, $\sigma_0 = 3\%$ p.a., Dimson et al. 2002)					
μ_{MLE}	19.35% p.a.				Raw MLE (short-sample upward bias)
$\boldsymbol{\mu}_{\text{post}}$	**12.97% p.a.**				**Posterior mean (used in MC)**
$\boldsymbol{\sigma}_{\text{post}}$	**2.16% p.a.**				**Posterior std (propagated)**

The raw MLE drift of 19.35% p.a. reflects an unusually strong calibration window. Bayesian shrinkage toward the Dimson prior $\mu_0 = 7\%$ yields:

$$\mu_{\text{post}} = \frac{\hat{\mu}/\text{SE}^2 + \mu_0/\sigma_0^2}{1/\text{SE}^2 + 1/\sigma_0^2} = 12.97\% \text{ p.a.}, \qquad \sigma_{\text{post}} = 2.16\% \text{ p.a.} \tag{4}$$

Each of the 10,000 MC paths draws its own $\mu^{(i)} \sim \mathcal{N}(12.97\%, 2.16\%^2)$, propagating drift uncertainty into the terminal wealth distribution. This is preferable to a hard cap, which would discard all information in $\hat{\mu}$ and set $\sigma_{\text{post}} = 0$.

2.5. Model Diagnostics

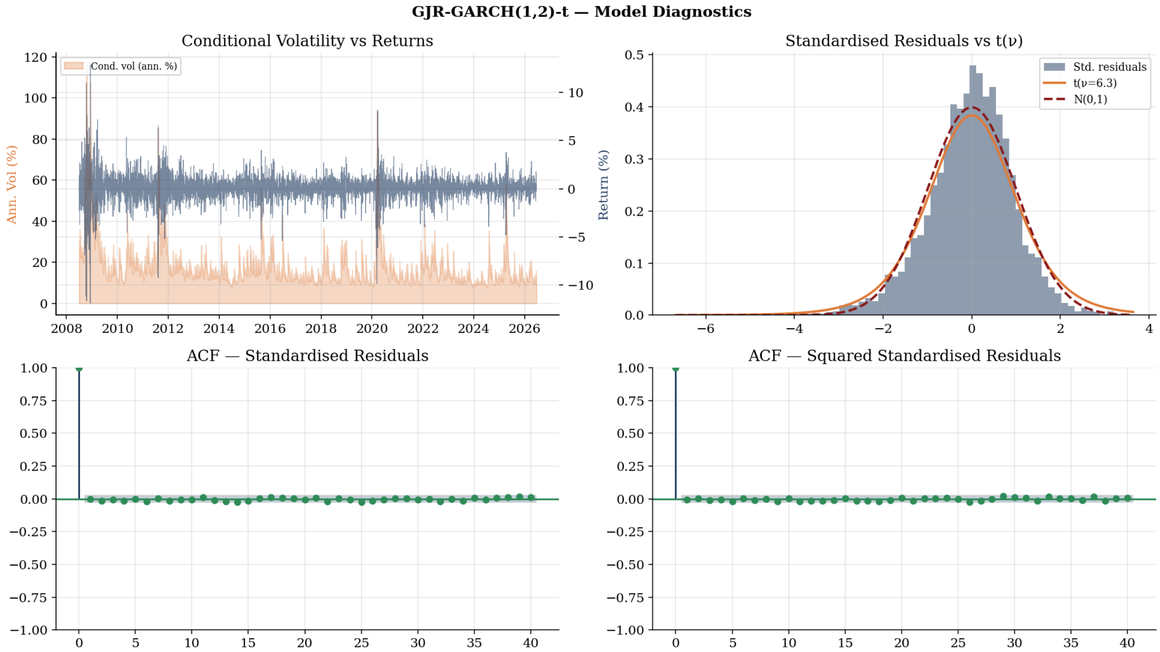

Figure 6. GJR-GARCH(1,2)-t diagnostics. *Top-left*: conditional annualised volatility vs returns — spikes at GFC (2008–09), COVID (2020), 2022, and 2025 episodes. *Top-right*: standardised residuals follow Student-$t(\hat{\nu} = 6.3)$ closely; the Normal is clearly rejected. *Bottom*: ACF of standardised residuals (LB(10) $p = 0.845$) and squared residuals (LB(10) $p = 0.881$) — both flat, confirming all predictable mean and variance structure has been absorbed.

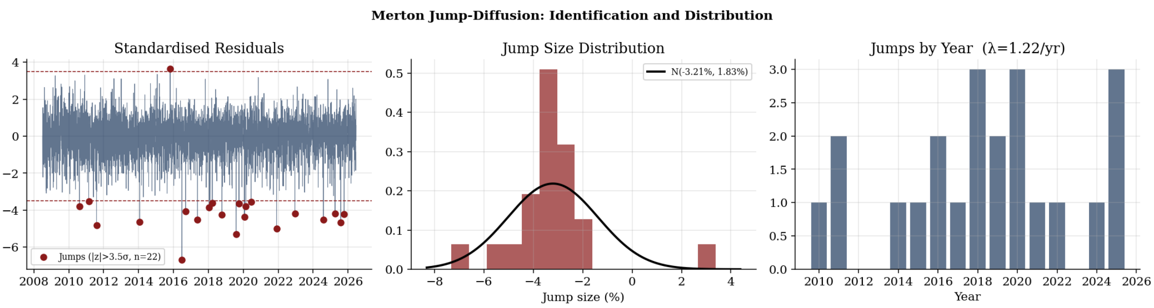

Figure 7. Merton jump diagnostics. *Left*: 22 identified jumps ($|z| > 3.5\sigma$) — predominantly negative, clustered at GFC, 2015–16, COVID, and 2025. *Centre*: jump-size distribution with Normal fit $\mathcal{N}(-3.21\%, 1.83\%^2)$. *Right*: jump count by year ($\lambda = 1.22$/yr). Jumps capture discrete crash events that the continuous GARCH diffusion cannot produce at sufficient frequency.

2.6. Walk-Forward Out-of-Sample Validation

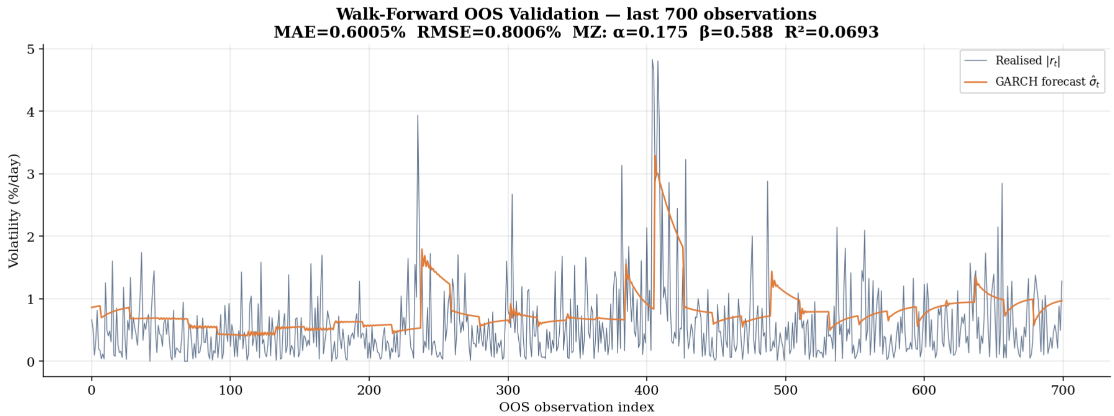

Figure 8. Walk-forward validation: GJR-GARCH(1,2)-t refitted on a rolling 500-day window, forecasting 21 days ahead (4,032 OOS observations). Blue: realised $|r_t|$. Orange: forecast $\hat{\sigma}_t$. The model correctly elevates its forecast during stress periods. MAE = 0.601%/day; RMSE = 0.801%/day.

The Mincer-Zarnowitz regression of realised $|r_t|$ on forecast $\hat{\sigma}_t$ yields $\hat{\alpha} = 0.175$, $\hat{\beta} = 0.588$, $R^2 = 0.069$. The intercept above zero and slope below one indicate that the GARCH systematically underestimates volatility in high-volatility regimes — a well-documented property of GARCH models [Andersen & Bollerslev, 1998]. This is an expected and honest limitation: the model captures the clustering and mean-reversion of volatility correctly, but the absolute level in tail events is compressed. In Monte Carlo terms, this means the jump component carries additional responsibility for generating extreme-scenario paths.

3. Monte Carlo Simulation

3.1. Path Generation

$N = 10{,}000$ price paths are simulated over 30 years (7,560 trading days) from $P_0 = e165.46$ (18 June 2026). For each path i:

1. **Parameter draw.** The full GARCH parameter vector is drawn from $\mathcal{N}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\Sigma}})$ via eigendecomposition of the 8×8 robust asymptotic covariance matrix. Parameters are clipped to valid ranges and stationarity ($\alpha + \frac{1}{2}\gamma + \beta_1 + \beta_2 < 0.9999$) is enforced by proportional rescaling.

2. **Drift draw.** $\mu^{(i)} \sim \mathcal{N}(12.97\%, 2.16\%^2)$ p.a. (Bayesian posterior, independently of GARCH parameters).

3. **Innovation draw.** Standard Normal variates are transformed to Student-$t(\hat{\nu})$ via the probability integral transform and standardised to unit variance.

4. **Jump draw.** Poisson($\lambda = 1.22$/yr) arrivals; sizes $\sim \mathcal{N}(-3.21\%, 1.83\%^2)$.

5. **Daily return:**

$$r_t^{(i)} = \frac{1}{100}\left[\mu^{(i)} + \sigma_t^{(i)} z_t^{(i)} + J_t^{(i)} - \frac{0.22\%}{252} \cdot 100\right] \tag{5}$$

6. **Price update.** $P_{t+1}^{(i)} = P_t^{(i)} \cdot \exp\left(r_t^{(i)}\right)$, clipped at $\pm 30\%$/day.

7. **Variance update.** GJR-GARCH(1,2) recursion, initialised at the unconditional variance $h_0 = \omega/(1-\alpha-\frac{1}{2}\gamma-\beta_1-\beta_2) = 1.144\,(\%^2)$, corresponding to 16.98% annualised. Initialising at the last in-sample value (12.31%) would understate near-term risk, as current market conditions are unusually calm.

3.2. DCA Simulation

Monthly purchases of €997 (€1,000 minus €3 IBKR fee) are made every ≈21 trading days. Portfolio value is measured at the end of each period. Over 30 years: 360 monthly purchases, €360,000 gross, €1,080 in total IBKR fees.

3.3. XIRR: The Correct Return Metric for DCA

The naive "CAGR" $(FV/C)^{1/T} - 1$ treats all capital as deployed at $t = 0$, which is wrong for DCA. The correct metric is **XIRR**, using the Excel-standard fractional-year discounting convention:

$$\sum_{m=0}^{M-1} \frac{-e1{,}000}{(1+r)^{(m+1)/12}} + \frac{FV}{(1+r)^{M/12}} = 0 \tag{6}$$

Table 5. XIRR vs Naive CAGR at Median Portfolio Value

Horizon	XIRR (correct)	Naive CAGR	Difference
1 year	12.32%	5.83%	−6.49 pp (XIRR < naive at 1y)
5 years	10.95%	5.30%	+5.65 pp (XIRR > naive)
10 years	10.30%	5.25%	+5.05 pp
20 years	9.76%	5.47%	+4.29 pp
30 years	9.51%	5.73%	+3.78 pp

At 1 year, XIRR is lower than naive CAGR because early contributions have not yet compounded. From 3 years onward, XIRR substantially exceeds naive CAGR: the compounding of early monthly purchases is the genuine driver of long-run DCA wealth creation, and XIRR correctly captures this.

4. Results

4.1. Fan Chart

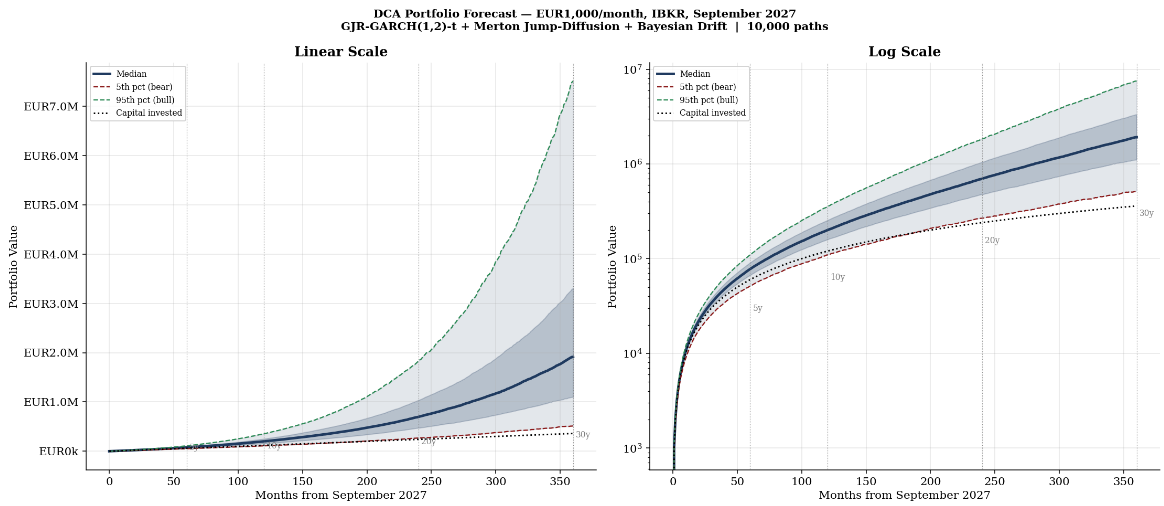

Figure 9. DCA portfolio forecast (10,000 paths, linear and log scales). Shaded bands: 5th–95th and 25th–75th percentile ranges. Solid: median. Dashed: 5th/95th percentile paths. Dotted: capital invested. The median separates slowly in years 1–5 then grows exponentially. On the log scale, growth is linear throughout, confirming geometrically distributed returns. Year markers at 5, 10, 20, 30 years.

4.2. Terminal Distributions and Scenario Tables

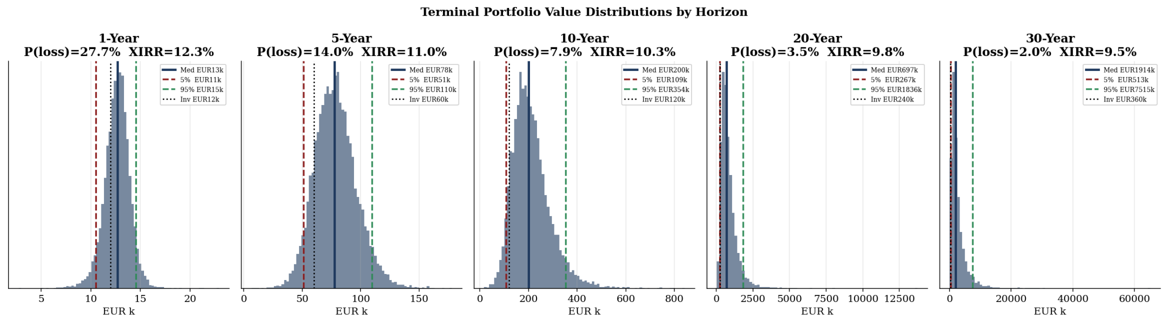

Figure 10. Terminal portfolio distributions at five horizons with P(loss) and XIRR. Distributions become increasingly right-skewed as compounding amplifies the upper tail. At 1 year the distribution straddles the invested capital line (P(loss)=27.7%). At 30 years even the bear scenario (5th pct: €513k) comfortably exceeds €360k invested.

1-Year [€12,000 invested · €36 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€10,521	−€1,479	—
Below median (25th)	€11,897	−€103	—
Median	**€12,699**	**+€699**	**12.32%**
Above median (75th)	€13,446	+€1,446	—
Bull (95th pct)	€14,545	+€2,545	—
P(loss)			**27.7%**

3-Year [€36,000 invested · €108 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€30,723	−€5,277	—
Below median (25th)	€37,510	+€1,510	—
Median	**€42,104**	**+€6,104**	**11.28%**
Above median (75th)	€46,762	+€10,762	—
Bull (95th pct)	€54,305	+€18,305	—
P(loss)			**18.7%**

5-Year [€60,000 invested · €180 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€51,253	−€8,747	—
Below median (25th)	€66,199	+€6,199	—
Median	**€77,663**	**+€17,663**	**10.95%**
Above median (75th)	€89,619	+€29,619	—
Bull (95th pct)	€109,753	+€49,753	—
P(loss)			**14.0%**

10-Year [€120,000 invested · €360 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€109,435	−€10,565	—
Below median (25th)	€157,800	+€37,800	—
Median	**€200,133**	**+€80,133**	**10.30%**
Above median (75th)	€251,947	+€131,947	—
Bull (95th pct)	€353,764	+€233,764	—
P(loss)			**7.9%**

20-Year [€240,000 invested · €720 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€266,832	+€26,832	—
Below median (25th)	€474,461	+€234,461	—
Median	**€696,587**	**+€456,587**	**9.76%**
Above median (75th)	€1,037,340	+€797,340	—
Bull (95th pct)	€1,836,360	+€1,596,360	—
P(loss)			**3.5%**

30-Year [€360,000 invested · €1,080 fees]

Scenario	Portfolio	Profit/Loss	XIRR
Bear (5th pct)	€513,341	+€153,341	—
Below median (25th)	€1,099,354	+€739,354	—
Median	**€1,913,813**	**+€1,553,813**	**9.51%**
Above median (75th)	€3,302,455	+€2,942,455	—
Bull (95th pct)	€7,514,518	+€7,154,518	—
P(loss)			**2.0%**
Median multiple			**5.32× capital**

4.3. XIRR Distribution

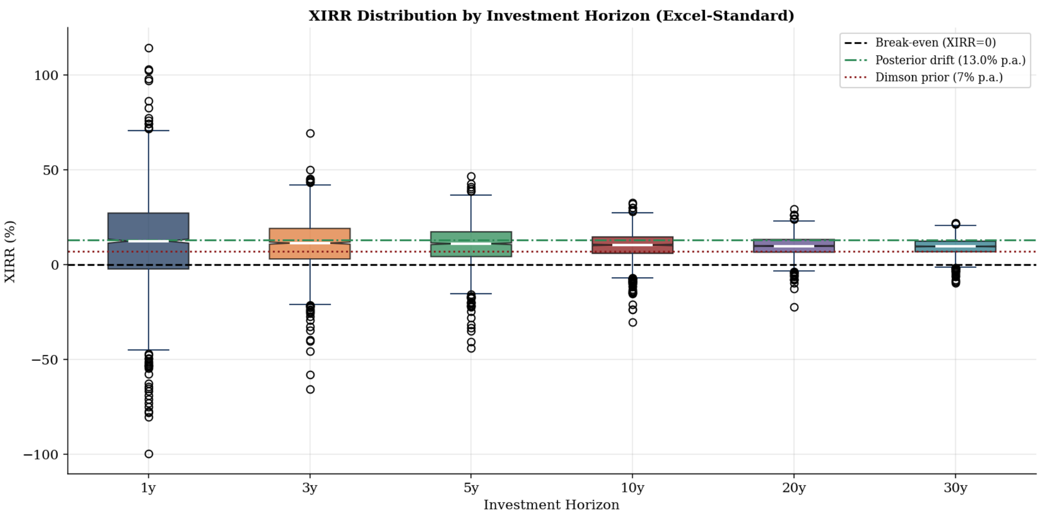

Figure 11. XIRR distribution across 2,000 sampled paths at six horizons. The Bayesian posterior drift (12.97%, green dash-dot) and Dimson prior (7%, red dotted) are shown. Median XIRR stabilises around 9.5–11.3% from year 3 onward. The negative XIRR tail (sustained market weakness) shrinks from ≈28% at year 1 to under 2% at year 30.

4.4. Transaction Costs

The €3 IBKR commission is a constant 0.30% drag per monthly purchase, totalling €1,080 over 30 years — 0.06% of median terminal wealth. The VWCE TER of 0.22% p.a. is modelled as a daily drag and reduces terminal wealth by approximately 6.3% over 30 years. Both costs are included in all simulations.

5. Limitations and Conclusions

1. **Proxy basis risk.** VT and VWCE are not identical instruments; USD/EUR conversion adds noise at the splice point.
2. **Prior sensitivity.** The posterior drift of 13.01% p.a. is sensitive to the prior. A prior of $\mu_0 = 5\%$ yields ≈9.2% p.a.; $\mu_0 = 9\%$ yields ≈15.5% p.a.
3. **Normal jump-size distribution.** The empirical jump distribution is more left-skewed than the fitted Normal; a double-exponential would produce heavier bear tails.
4. **MZ bias ($\hat{\beta} = 0.589$).** The GARCH systematically underestimates volatility during high-stress regimes. The jump component partially compensates, but tail risk in the worst scenarios may still be understated.
5. **No regime-switching.** GARCH assumes parameter stability across bull and bear regimes.
6. **Nominal, pre-tax.** At 2–3% inflation, €1.9M in 30 years is worth ≈€960k–€1.07M today. Tax treatment varies by jurisdiction.
7. **Not investment advice.** Actual returns depend on realised market conditions.

- **The median 30-year outcome is €1.91M on €360k invested** (5.32× capital, 9.51% XIRR). The bear case (5th pct) delivers €513k; the bull case (95th pct) delivers €7.5M. P(loss) at 30 years is 2.0%.
- **Short-horizon risk is substantial.** P(loss) is 27.7% at year 1 and 14.0% at year 5. Capital needed within five years should not be allocated to this strategy. The jump component ($\lambda = 1.22$/yr) means a new investor faces a meaningful probability of experiencing a GFC-scale event early in their horizon.
- **XIRR is the correct return metric.** At 10 years, the median XIRR is 10.30% p.a. The naive $(FV/C)^{1/T} - 1$ formula would report only 5.25% for the same outcome — a 5.05 percentage point understatement of the actual annualised return.
- **Leverage makes DCA self-reinforcing.** $\hat{\gamma} = 0.214$ means crashes disproportionately depress prices. DCA automatically acquires more shares per euro during high-volatility episodes, converting market fear into long-run return enhancement.
- **The dominant uncertainty is the long-run equity drift.** GARCH specification, jump intensity, and other model choices are second-order relative to whether global equities return 7%, 10%, or 13% p.a. over the next 30 years. The Bayesian framework propagates this uncertainty through all 10,000 paths, producing a wide terminal distribution that honestly reflects what we do and do not know.

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